10 TACTICS, MANIFESTATIONS, CAPABILITIES & GUEST LECTURE

ESSENCE – CHAPTERS 15, 16 & 17 (SLIDES ONLY)

GUEST LECTURE: KLAUS BUNDVIG, NETCOMPANY

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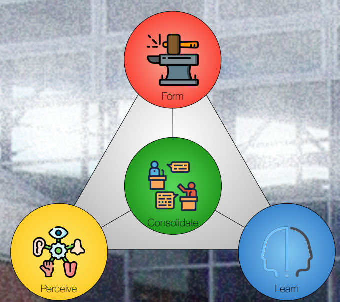
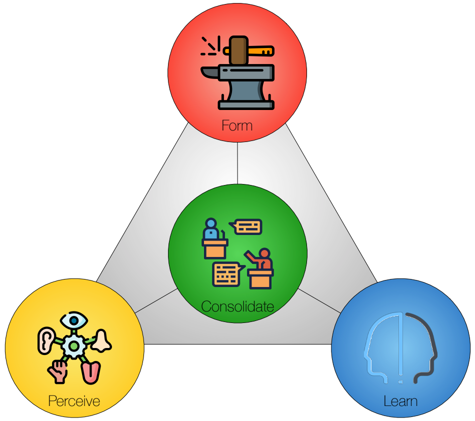
SWI 2026



Agenda

- Exercise 9:
  - Evolvability & General Innovation Theory.
- Lecture 10:
  - Chapter 15 in Essence: Tactics (slides only).
  - Chapter 16 in Essence: Manifestations (slides only).
  - Chapter 15 in Essence: Capabilities (slides only).
  - Guest lecture: Klaus Bundvig, Netcompany.

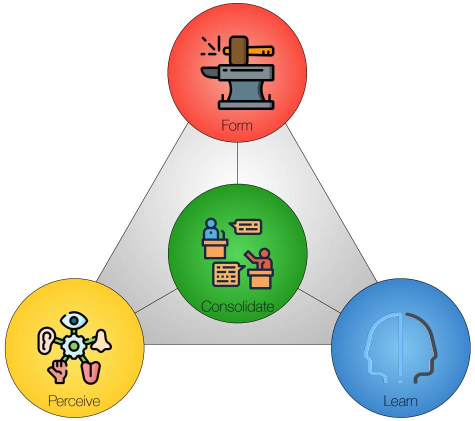


Follow-up: Exercise 9

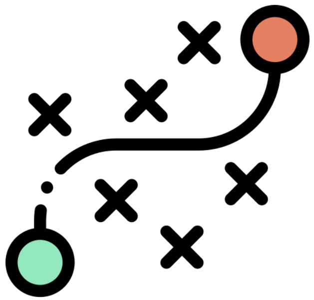
Exercise 9 (Evolvability & General Innovation Theory)

This exercise concerns the evolvability and innovation characteristics of your project.

- Describe and illustrate diffusibility in relation to your project.
- Describe and illustrate adoptability in relation to your project.
- Evaluate your project using one or more of the evaluation types presented on the slides.
- Describe your project using the terms invention, innovation, exploitation, and diffusion.
- Describe the novelty, radicality, and timing of your project.



Tactics

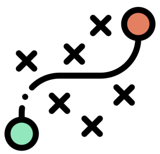


Do the difficult things while they are easy and do the great things while they are small.
A journey of a thousand miles must begin with a single step
Lao Tzu (6th century BC–5th century BC)

Three Levels of Abstraction

Rationale – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.

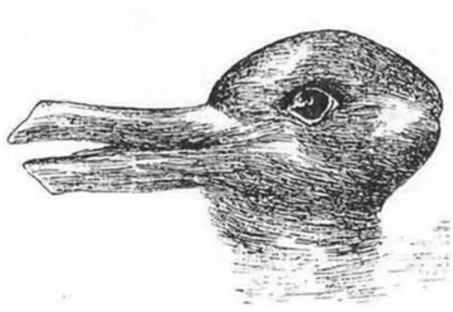
Strategy – the overall plan for addressing the problem, including its scope and context, the essential parts needed to build the solution, and any limitations or constraints that might influence its feasibility. At the strategy level, we also evaluate the potential for future growth.

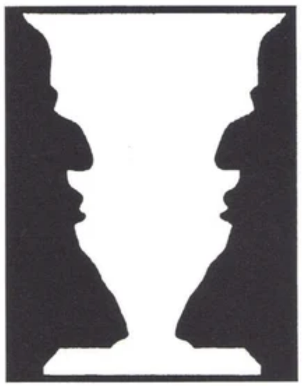
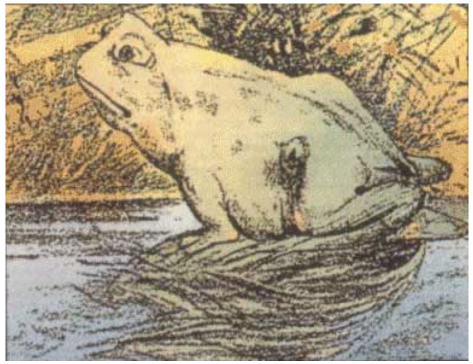


Tactics – actions planned in accordance with the strategy to achieve specific ends.

Observation is Theory-laden

- Norwood Russell Hanson (1924 – 1967) was an American philosopher of science.
- He argued that observation is theory-laden. Observation language and theory language are deeply connected.
- What we perceive is not what our senses receive, but is filtered through our existing preconceptions.
- We cannot separate sensing and interpreting.
- For that reason, we see the abstract and the concrete as complementary in inquiry and resolution.



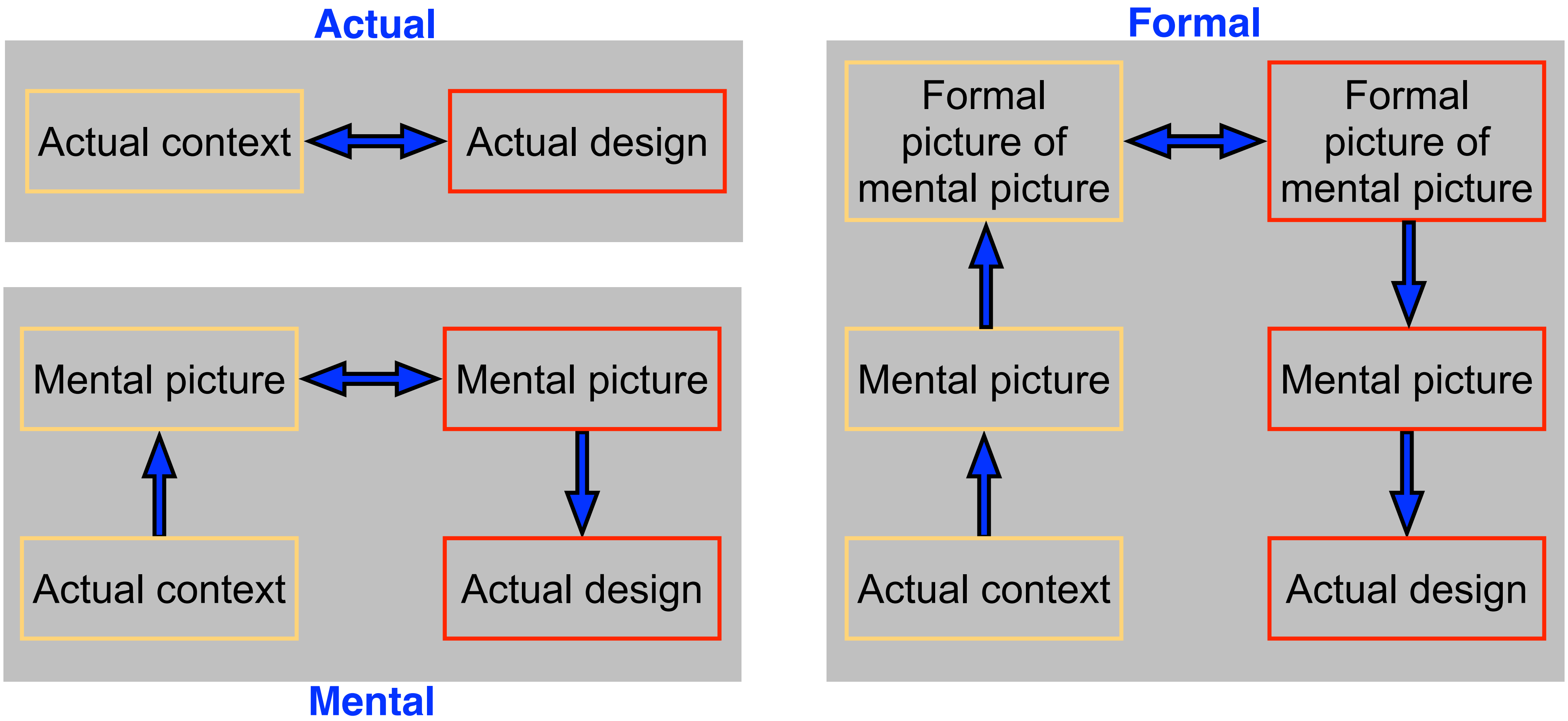
Principles of problem formulation and solutions

- The philosophical foundation is Deweyan pragmatism, drawing on terms such as transaction, inquiry, and situation.
- The work sequence can begin with identifying the problem and then finding a solution, or start with a solution and determine appropriate problems.
- The purpose of the problem formulation principles is to strike a balance between the concrete and the abstract.
- Similarly, the purpose of the solution principles is to strike a balance between the concrete and the abstract.
- A concrete formulation of a problem helps develop solutions that can be tested and confirmed.
- An abstract formulation of a problem helps develop solutions that are general and can be adjusted to address problems that are similar, in whole or in part.
- At the concrete level, the problem manifests as a set of manifestations.
- Similarly, at the concrete level, the solution is detailed in terms of capabilities.

Problem and solution formulation are linked

- Moving between the concrete and the abstract.
- Moving between the problem side and the solution side.
- Moving between the current situation and future options.

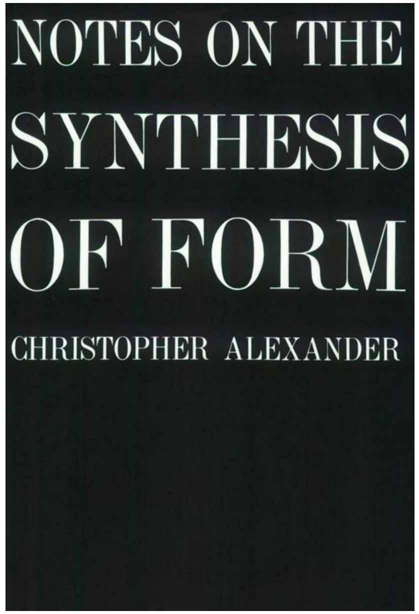
Actual, Mental, and Formal Views on Situations and Designs



Inspired by Alexander, C. (1964). *Notes on the synthesis of form* (Seventh printing, 1973 ed.). Cambridge, Mass: Harvard University Press.

The Unselfconscious, the Selfconscious, and the Qualified

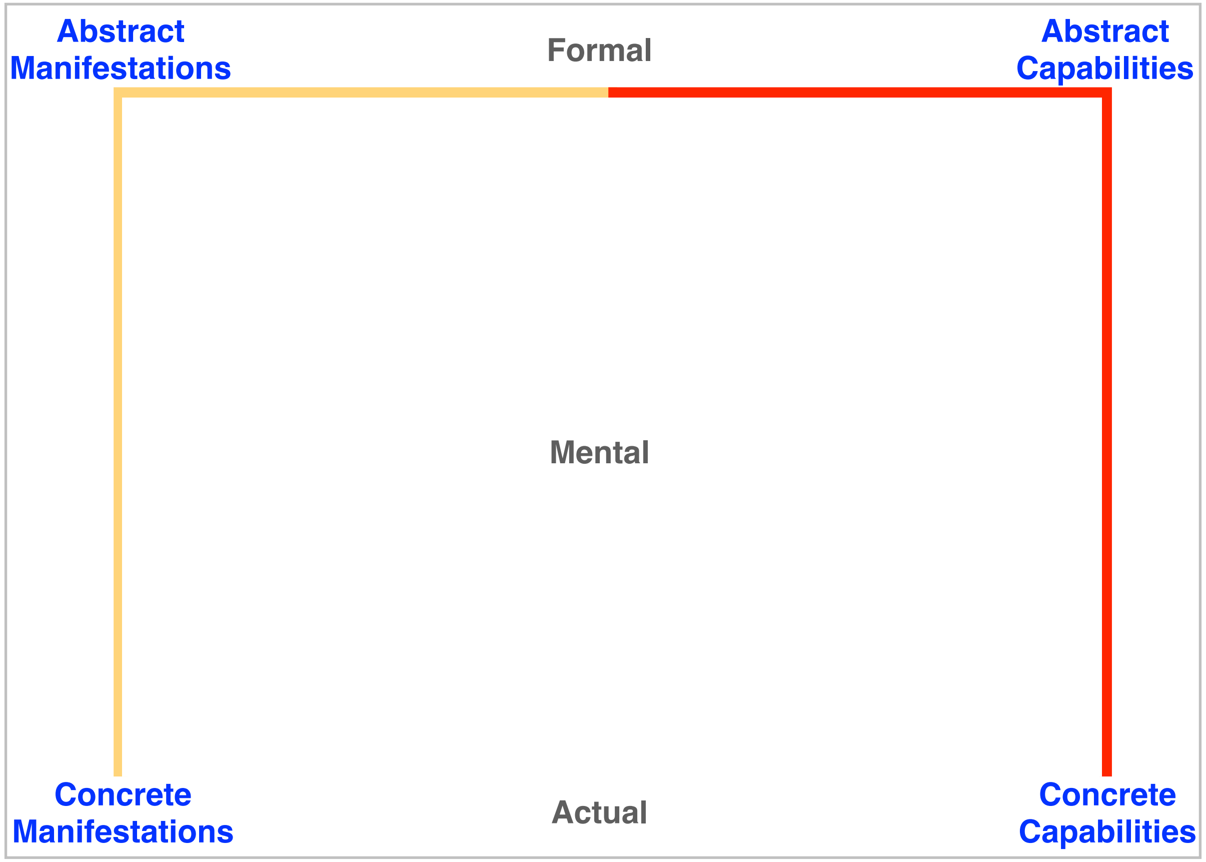
- The unselfconscious "designer" works only at the level of the actual. The problem-solver simply reacts to misfits by changing them.
- The selfconscious designer works also at the mental level. Design is shaped intuitively by a conceptual interaction between a mental picture of the context and a repertoire of ideas.
- The qualified designer also works on the formal level. Design is shaped by selecting patterns based on their pros and cons.



First out 1964

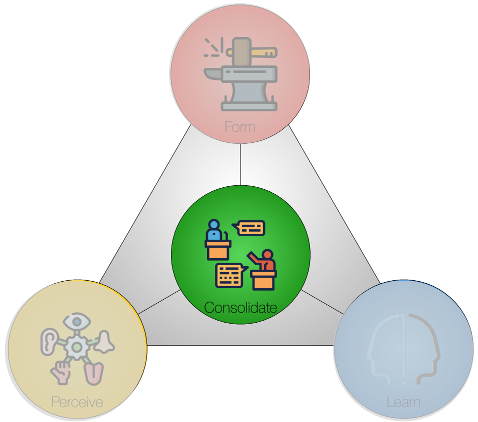
Inspired by Alexander, C. (1964). *Notes on the synthesis of form* (Seventh printing, 1973 ed.). Cambridge, Mass: Harvard University Press.

The abstract and the concrete are complementary

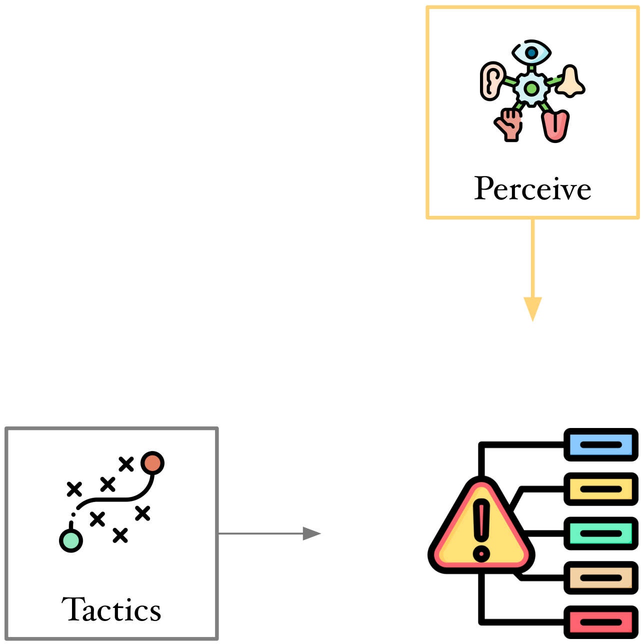


Inquiry is a movement between the situation, the problem, and the solution.
Push or Pull?—Inquiry may begin on either the problem or the solution side.

With inspiration from Jes Adolphsen and Finn Kjærsdam



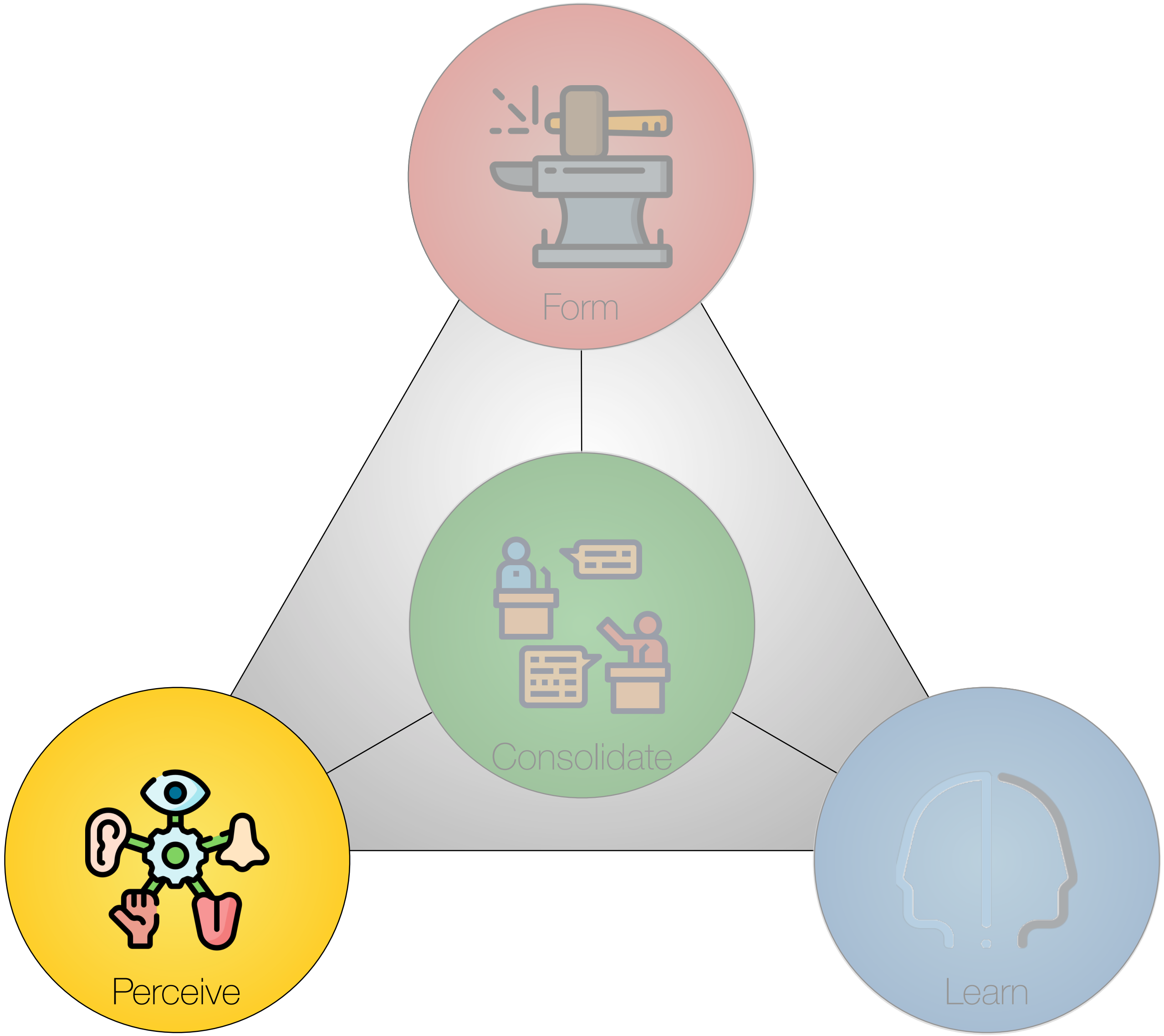
Manifestations



Solving a problem simply means representing it so as to make the solution transparent.
Herbert Simon (1916–2001)

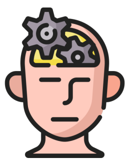
Four Core Activities

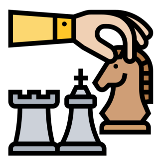
- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.

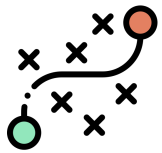


Icons from Flaticon.com

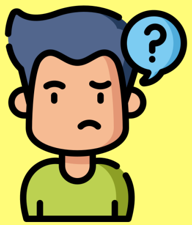
Perception Terminology







Problem



We see a problem as a situation to be handled, where problems appear as manifestations.

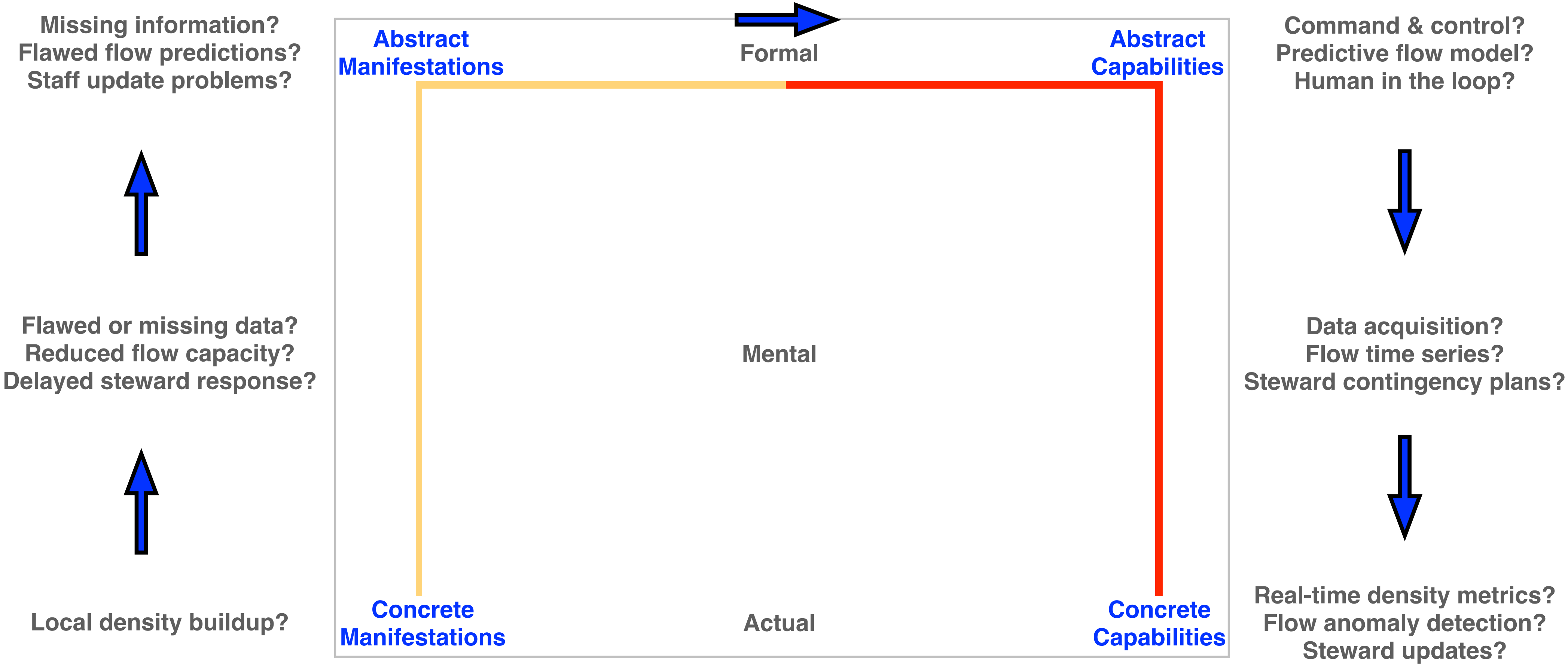
Outer Environment

The problem is found in the outer environment. The design is adapted to this domain.

Manifestations

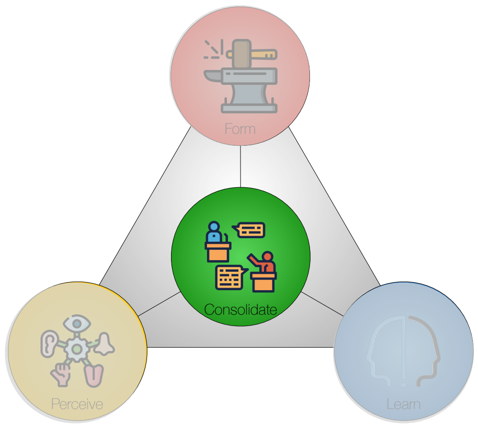
Manifestations are objects, actions, or events that reflect or embody the problem.

Concrete vs. Abstract – Demand Pull

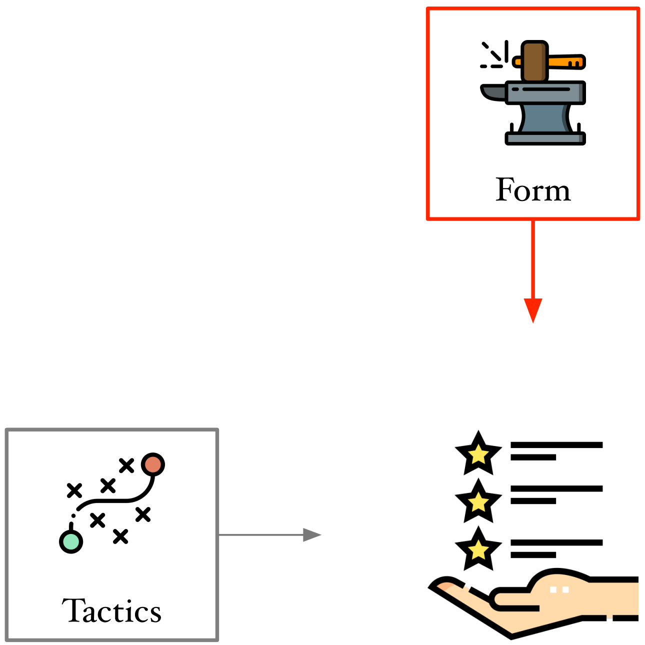


ClusterDetect illustration

Pull—from problem toward solution. Making decisions on the way.
A problematic situation is noticed first. Designers identify manifestations, formulate the problem, and then develop a solution.



Capabilities



We define organic order as the kind of order that is achieved when there is a perfect balance between the needs of the parts, and the needs of the whole.
Christopher Alexander (1936–2022)

Four Core Activities

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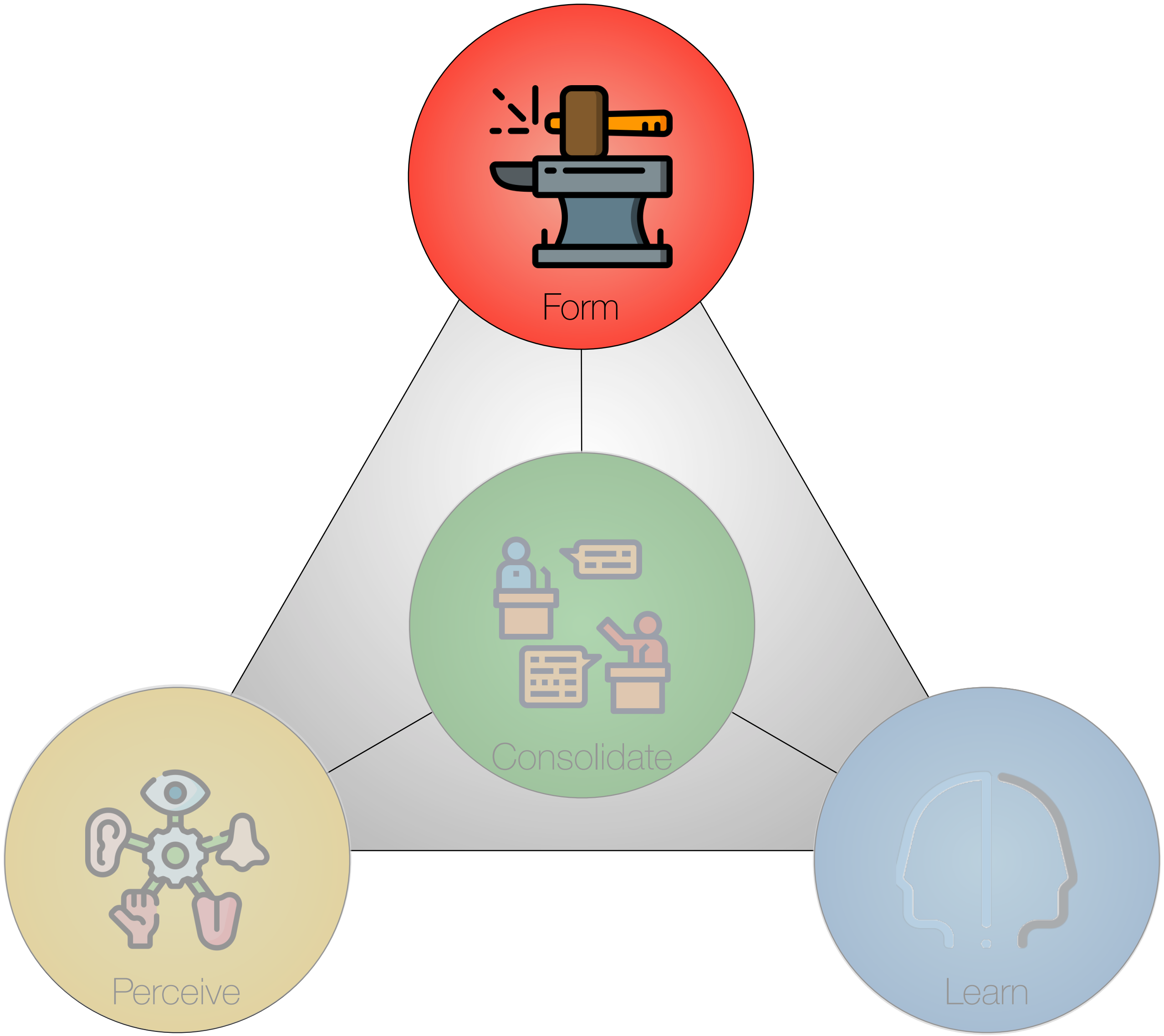
- *Learn* – appraising the design.



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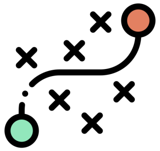
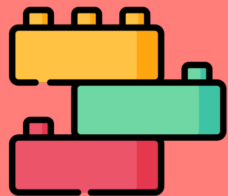
Form Terminology



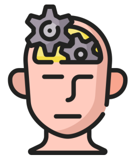
Capabilities

Capabilities are the features that are designed to address manifestations.

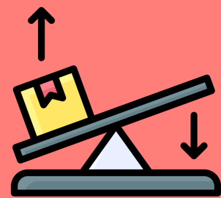
Inner Environment



The inner environment is what provides the capabilities.

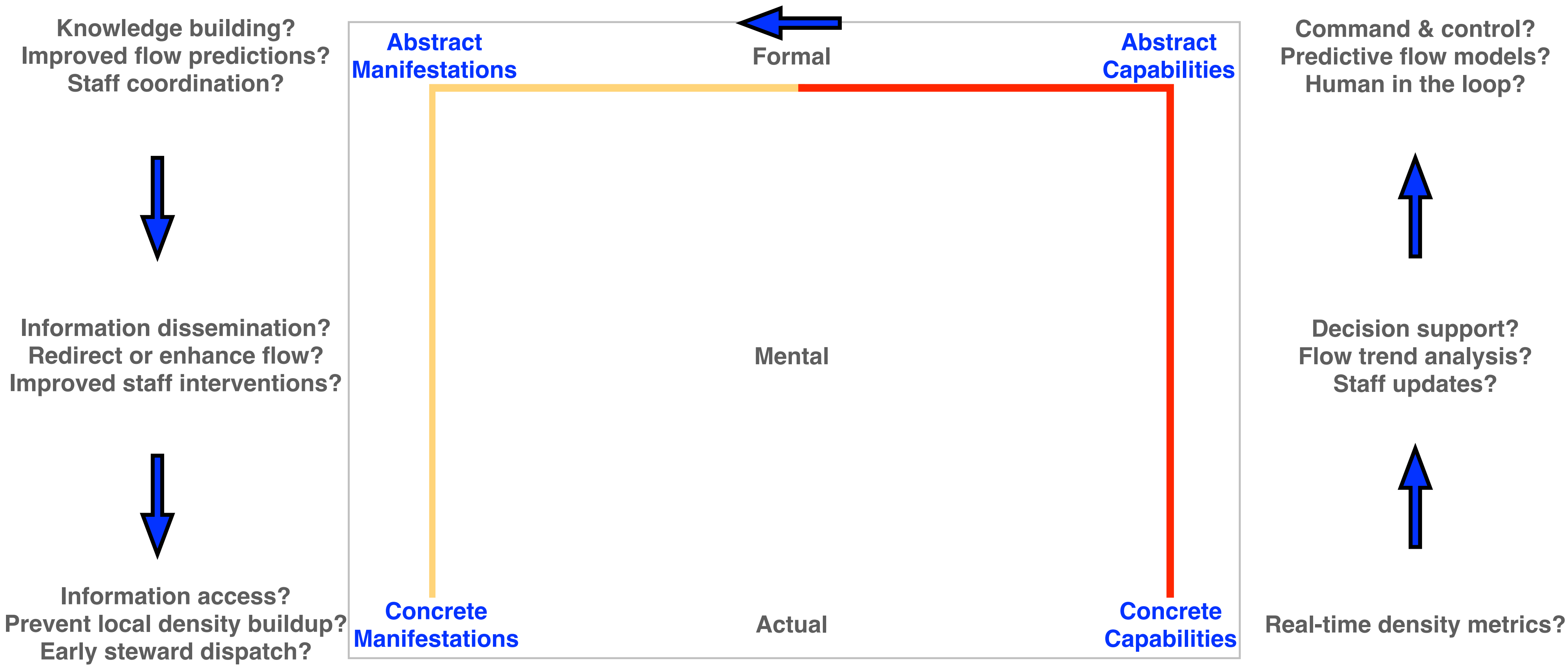


Leverage



Leverage points allow for the most valuable capabilities.

Concrete vs. Abstract – Technology Push



ClusterDetect illustration

Push—from solution toward problem

A technological possibility appears first. Designers explore what kinds of problems it may address, under what conditions, and for whom. What problematic situations could this help transform?

Problem-solving by Abstraction

- Patterns (Alexander, Gamma et al., Rising).
- Metaphor use (Dalsgaard, Beck).
- Innovation Games (Hohmann).
- Lateral thinking (De Bono).
- SCAMPER (Eberle).
    - Substitute, combine, adjust, modify, put to other uses, eliminate, reverse.
- TRIZ (Altshuller)
- and many more.

Patterns

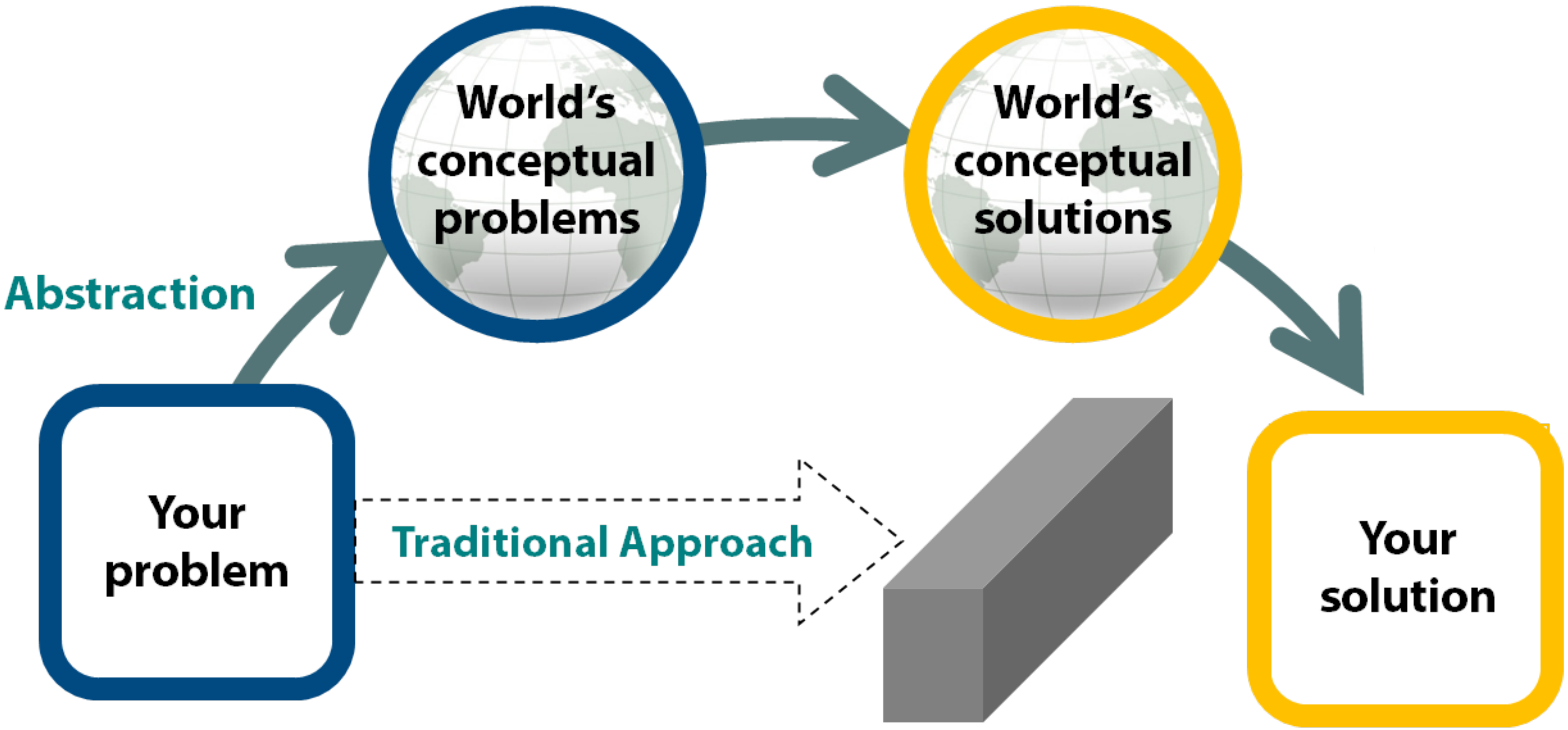
- Alexander (Architecture and general design).
- Gamma et al (Software design patterns).
- Rising (Application development).

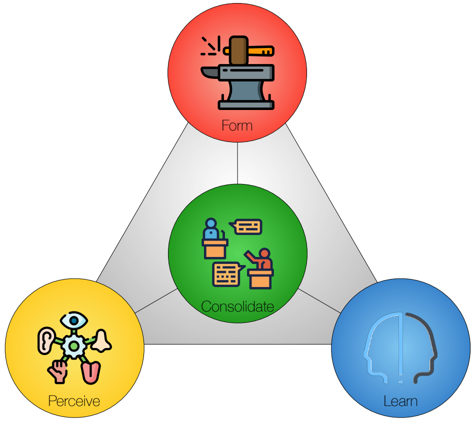
40 TRIZ Principles

1 Segmentation	2 Taking out	3 Local quality	4 Asymmetry
5 Merging	6 Universality	7 Nested doll	8 Anti-weight
9 Preliminary anti-action	10 Preliminary action	11 Beforehand cushioning	12 Equipotentiality
13 The other way round	14 Spheroidality - Curvature	15 Dynamics	16 Partial or excessive acti...
17 Another dimension	18 Mechanical vibration	19 Periodic action	20 Continuity of useful acti...
21 Skipping	22 *Blessing in disguise* o...	23 Feedback	24 Intermediary
25 Self-service	26 Copying	27 Cheap short-living obje...	28 Mechanics substitution
29 Pneumatics and hydraul...	30 Flexible shells and thin f...	31 Porous materials	32 Color changes
33 Homogeneity	34 Discarding and recoveri...	35 Parameter changes	36 Phase transitions
37 Thermal expansion	38 Strong oxidants	39 Inert atmosphere	40 Composite materials

https://www.triz40.com/aff_Principles_TRIZ.php

TRIZ Flowchart





Next Time

Next Exercise and Lecture

- Exercise 10:
  - Tactics, Manifestations & Capabilities.
- Lecture 11:
  - Chapter 18 in Essence: Merit

Exercise 10 (Tactics, Manifestations & Capabilities)

This exercise concerns the tactics, manifestations, and capabilities of your project.

- Will you characterize your approach as push or pull?
- Describe concrete and abstract manifestations.
- Describe concrete and abstract capabilities.
- Will your abstractions improve evolvability and make your design future-proof? How?